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Options for Wireless Development

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The Ideal Wireless Solution is...

- A native Palm OS application
- with wireless and cable connectivity
- directly to the server
- with full encryption.

Agenda

- Palm’s commitment to wireless
- Putting the user first
- Comparison of development methods

Three routes to the Mobile Internet

- Integrated wireless devices
  - Data-centric
  - Data and voice
- Leveraging expansion for the connection
  - Snap-on peripherals
  - Plug-in expansion cards
- Two piece solution: Palm and cell phone
  - Mobile Internet Kit

Content is King

- We enable information access
  - Of any type
  - Using any protocol
- We promote and endorse only the best user experiences

Putting the User First

- A framework to compare development methods
- Using three riddles

Riddle One

Riddle One

- **Question:**
  What is the best way to get from San Francisco to Boston?
  - Hint: Travel time
  - Plane: 6 hours
  - Car: 6 days
  - Boat: 6 weeks



- **Answer:**
  Depends on what you are carrying
  - Small bag: take the plane
  - Small couch: drive a car
  - Small load of grain: send via boat

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Lesson One: Data Transfer

- Consider data characteristics
  - Size
  - Volatility
  - Value
- Leverage any and all connections INTELLIGENTLY
  - Serial or ethernet cradle
  - IR stations
  - Wireless: (fast) fixed, cellular, paging, Bluetooth
  - Wireline

Riddle Two

Riddle Two

- **Question:**
  When is air valuable?
  - Hint: Stop breathing RIGHT NOW!



- **Answer:**
  When it is immediately available
- Rejected Answers
  - When it is stored
  - When it is moved quickly

Lesson Two: Data Access

- Instant, consistent, reliable access
- Searching and cross-referencing with local databases
- Manipulation and operation processing with local processing
- Data is not valuable just because
  - there is lots of it
  - you can store hoards of it
  - you can push it quickly in many directions

Riddle Three

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Riddle Three

- **Question:**
  What is the best way to get a cat out of a tree?
  - Hint: You cannot kill the cat
- **Answer:**
  Open a can of cat food at the base of the tree, requiring:
  - Communication with the cat, using sight, sound, smell, voice and touch
  - Tools appropriate for the problem
- Rejected Answers
  - Chain saw
  - Signal flags



Lesson Three: Data Interaction

- Productivity requires the user both receives and interacts with data.
- Display using
  - Color screen
  - Graphics
- Input using
  - Direct touch
  - Pull-down menus
  - Check boxes
  - Graffiti
  - Keyboards: pop-up and accessory

Comparison of Options

- Using this user-centric framework
- Comparing:
  - A native Palm OS solution with wireless capability
  - Web clipping
  - WAP
  - Web browsing

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A Native Palm OS Application: The Optimal Solution

- Use all available data transfer channels intelligently
  - For all Palm OS devices, with (or without) any type of wireless connectivity
- Use device-side databases, processing power, and interactive elements
  - Access data even when connection fails
  - Integrate various types of data together
  - Integrate Graffiti, touch screen, etc.
- Net Libraries for C-based applications

Web Clipping: A Simple and Elegant Intermediate Step

- Ease of Development: HTML
- Fast and inexpensive data transfer
- Reconfigured flow of information
- Feature-rich experience

Web Clipping: An Example

Launcher



Query Application



Clipping



WAP: Important but Immature

- WAP (and WML)
  - Perfect for small amounts of data
  - Requiring no reply
  - Where value is in currency
- Difficulties
  - Several versions and still evolving
  - Feature-poor paradigm
- Palm's WAP Support
  - Neomar and AU Systems are "favorites"
  - Included in the MIK

Web Browsing: The Last Option

- Wrong Format
  - 640-480 color screen
  - Graphics-intensive
- Wrong User
  - For a sedentary person
  - For a few long sessions per day
  - Shotgun blast of content
- Wrong Connection
  - Requires fast connection: >56K
  - Assumes persistent connection

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Q&A

Other Pertinent Sessions

- Sessions 307 and 308L: Adding wireless capability to a .prc (Native Application)
- Session 305: Creating web clipping applications
- Session 204: New features of Web Clipping

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Conclusion

- **Question:**
  Where are you going right now?
- Hint: Use a native Palm OS app
  - Multiple data transfer methods
  - Instant access
  - Rich user interaction

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